3

The matrix E_{31} has a 2 in the (3,1) position, which was used to eliminate the entry in row 3, column 1 of matrix A. When multiplying the elimination matrices $E_{32}E_{31}E_{21}$, the entry 10 in E arises from combining the multipliers: 2 (from E_{31}) and -2 (from E_{32}) acting on previous eliminations. Specifically, during multiplication, $2 \times 1 + (-2) \times (-4) = 10$.

In $L = E^{-1}$, the -2 in the (3,1) position comes from the inverse of E_{31}, which reverses the elimination step by negating the original multiplier. Thus, E_{31}^{-1} has -2, contributing to L.

Answer:

The 10 in E results from combining elimination steps, and the -2 in L is the inverse multiplier from E_{31}.

- $E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix}$,

4

$$\begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix}$$

$$\begin{matrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 4 & 0 \end{matrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ 16 \end{bmatrix}$$

$y = \frac{-4+5}{2} = \frac{1}{2}$

$2z = 10$

$\therefore z = 5$

$x = \frac{1}{2}$

add row 1 to 3 **then** add row 3 to 1.

$$\overset{1,3}{\left[\begin{pmatrix}1\\0\\1\end{pmatrix}+\begin{pmatrix}0\\1\\0\end{pmatrix}+\begin{pmatrix}0\\0\\1\end{pmatrix}\right]} = \overset{3!}{\begin{bmatrix}2&0&1\\0&1&0\\1&0&1\end{bmatrix}}$$

(11) $\begin{bmatrix}1&1&1\\2&1&0\\4&0&1\end{bmatrix}$

(12) $P=\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}\begin{bmatrix}$

$P^{-1}=P^T \quad P^TP=I \;, P^{-1}P=I$ (P are orthogonal Matrices)

(14) $\begin{bmatrix}1&a\\0&-1\end{bmatrix}\begin{bmatrix}1&a\\0&-1\end{bmatrix}\begin{bmatrix}1&a-a\\0&1\end{bmatrix}$

(15) a) $AB = AC$

$A^{-1}AB = AA^{-1}C \quad \therefore B=C$

b) $A=\begin{bmatrix}1&1\\\end{bmatrix}$ $\det=0$ so A non-

b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ invertible

$AB = AC \therefore AB - AC = AC - AC$

$\therefore A(B-C) = 0$ so $B-C$ is in nullspace of A.

Let, $B - C \begin{bmatrix} x & y \\ z & w \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} \Rightarrow \begin{bmatrix} x+z & y+w \\ x+z & y+w \end{bmatrix} = 0$$

$x + z = 0 \quad \therefore x = -z$

$y = -w$

$$\therefore \begin{bmatrix} -z & -w \\ z & w \end{bmatrix} \text{ or } \begin{bmatrix} x & y \\ -x & -y \end{bmatrix}$$